Architecting Identity: Authentication and Authorization for AI Agents

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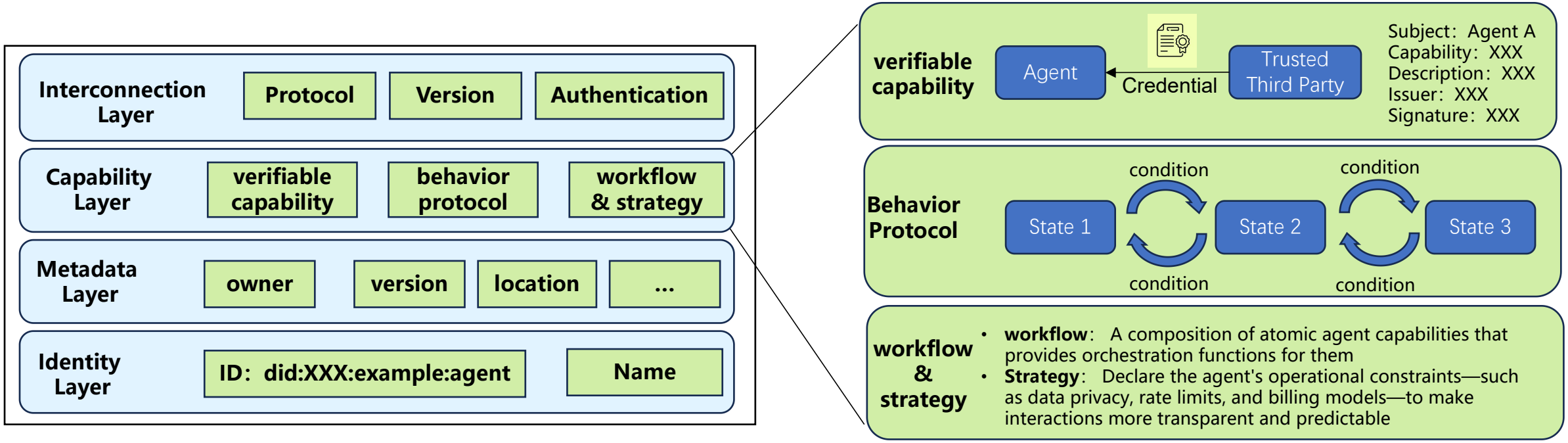
March 2026, IETF 125 @ Shenzhen

Key Issue 1: A unified framework for agent identity and capability description

- **Lack of Unified Standards**: The industry currently utilizes various agent identity and description schemes, Consequently, an agent "profile" (or "business card") generated on one platform may not be recognized or understood by another
- **Semantic Ambiguity**: Current descriptions of agent capabilities rely heavily on natural language and require interpretation by Large Language Models (LLMs), This diminishes the reliability and determinism of agent collaboration, particularly in scenarios that demand high precision and unambiguous execution, such as industrial production and financial transactions
- **Inadequate Representation of Dynamic and Contextual Capabilities**: Current solutions, such as Agent Cards, typically describe a static, stateless list of tools. They fail to express more complex capability logic, including state dependencies and capability composition.



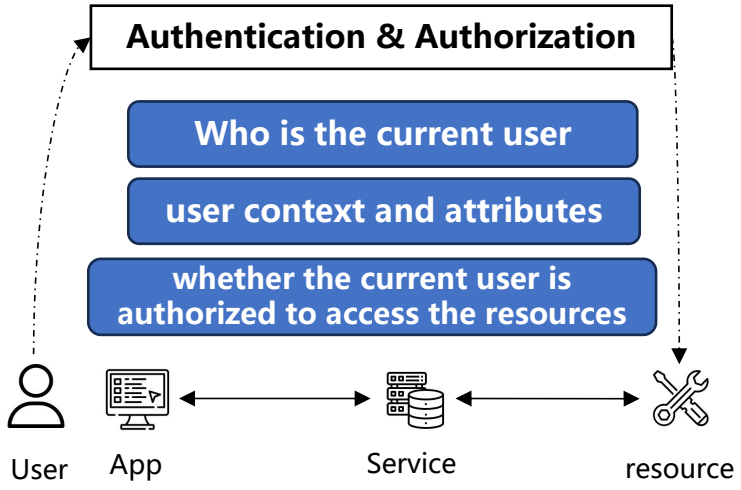
Key Issue 2: Multi-Agent Collaborative Authentication and Authorization Management

- How to ensure that the authentication and authorization mechanisms for AI Agents are trustworthy and secure
- How to effectively propagate and verify identity information, and execute authorization operations, across complex agent invocation chains

User-centralized

Authorization Point

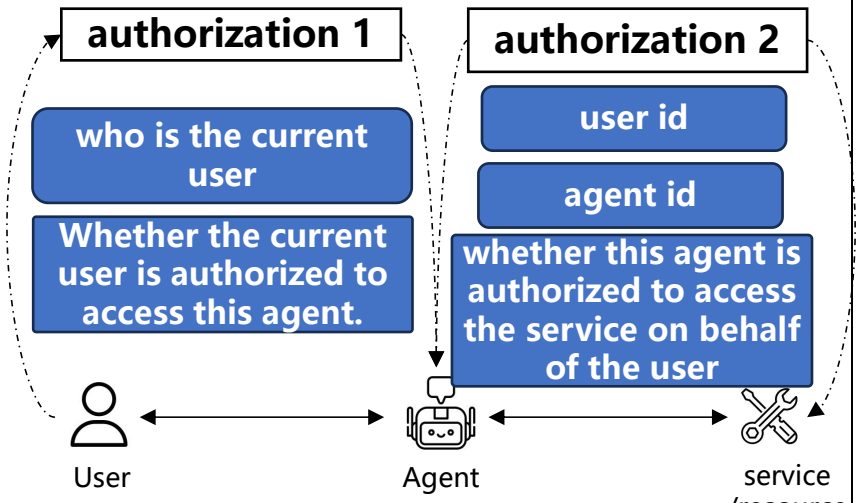
Each user acts as an independent point of authentication and authorization, where access control and authorization are enforced based on their identity, attributes, and current security context



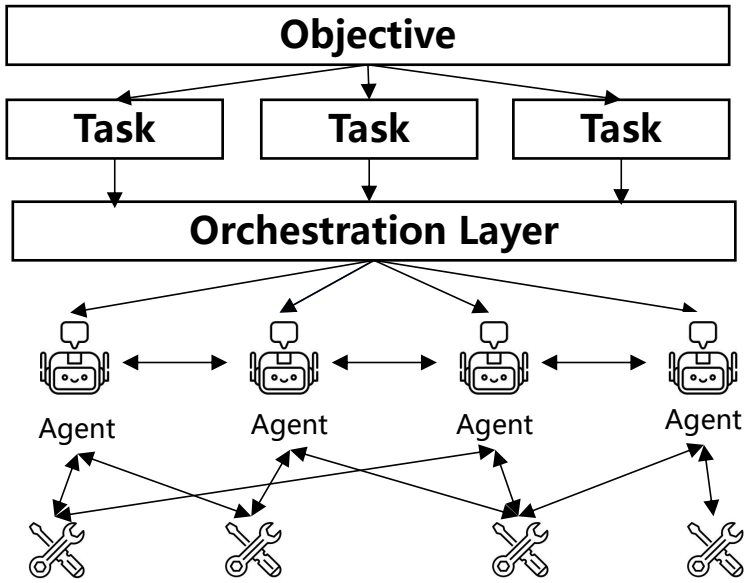
User-agent combination

Authorization chain

The user, the agent, and the resource/service collectively form an authentication and authorization chain



Multi-agent collaboration

Authorization mesh

In multi-agent collaboration scenarios, complex invocation relationships are formed between agents, as well as between agents and tools, constituting an authorization mesh.